# Supplementary Material for SG-ALNS: Segment-Level Graph RL for ALNS Configuration in the VRPTW

Duong Q. Vo
*Faculty of Information Technology, Vinh University*
Nghe An, Vietnam
235748020110196@vinhuni.edu.vn

Sang X. Tran*
*Institute of Information Technology and Digital Transformation, Vinh University*
Nghe An, Vietnam
sangtx@vinhuni.edu.vn

Uyen T. Nguyen*
*Faculty of Information Technology, Vinh University*
Nghe An, Vietnam
uyennt@vinhuni.edu.vn

## Implementation Details

*Instances and state graph.* With the depot horizon $H = l_0$, the six static node features are $x_i/100$, $y_i/100$, $d_i/Q$, $e_i/H$, $l_i/H$ and $\sigma_i/H$. The six dynamic features are computed on the *current* solution $x$ (not on the best one): waiting time, slack $l_i - b_i$ and forward time slack (Savelsbergh, 1992), each divided by $H$; cumulative load divided by $Q$; relative position in the route; and a routed flag (the depot has zeros). The edge set contains, for every customer, directed arcs to its 10 nearest customers, arcs between the depot and its 10 nearest customers, and both directions of every arc of the current routes (depot legs included). Each edge carries $c_{ij}/c_{\max}$ and a flag (0: neighbour arc, 1: route arc). The graph therefore grows with $n$, and mean and max pooling give an embedding of fixed size. The global vector holds $t/T$, $\tau/\tau_0$, the number of iterations since the last new best divided by $0.2T$ (capped at 1), $(f(x) - f(x^\star))/f(x^\star)$, $(f(x_0) - f(x^\star))/f(x_0)$ with $x_0$ the solution after vehicle reduction, the number of routes of $x$ divided by that of the I1 solution, and the success rate (share of uses that improved the current solution) of each of the nine operators in the last segment. SG-ALNS-C appends $n/200$ and $T/5{,}000$.

*Roulette wheel.* After each segment, $w_o \leftarrow \max\{(1-\rho)w_o + \rho\,\psi_o/u_o,\ 10^{-3}\}$, with $\psi_o/u_o := 0$ for an operator that was not used, so the weight of an unused operator decays geometrically towards the floor. Destroy and repair probabilities are the weights normalised within each group. Scores per use are 33 (new best), 9 (better than the current solution) and 13 (accepted worse solution). A repair that cannot insert every customer is rejected and scores 0.

*Factorised policy.* The three heads are categorical distributions over $a^1$ (19), $a^2$ (5) and $a^3$ (3) computed from the same embedding, i.e. conditionally independent given the state: $\log \pi(a \mid s) = \sum_k \log \pi_k(a^k \mid s)$, and the entropy used in the PPO loss is the sum of the three head entropies (exact for this factorisation). In an ablation, the removed head is masked to its expert value and contributes zero to both. The output weights are scaled by 0.1 at initialisation and the biases of the expert values are set to 3 ($a^1$, RW) and 2 ($a^2$, default range; $a^3$, $m = 1$); the neutral-prior variant sets all biases to 0.

*PPO.* Rewards are divided by a running standard deviation; each update runs 4 epochs over minibatches of 256 decisions, with value coefficient 0.5 and gradient-norm clipping at 0.5. Every 5 updates the greedy policy is scored against ALNS-RW on 24 fixed generated instances ($n = 100$); this validation is only logged, and the *final* checkpoint is evaluated. Training instances, validation instances (seed 12345) and the 24 tuning instances of the tuned baselines (seeds 90000–90023) are drawn from disjoint random streams of Solomon's generator; no benchmark instance is used for any selection.

*Objective and reward.* The best solution $x^\star$ is always tracked with $f = 10^4\mathrm{NV} + \mathrm{TD}$. Over all 17,772 runs of this study, the largest distance difference between two solutions of the same instance is 2,569 (the largest TD is 11,354), so $f$ orders every pair of our solutions lexicographically (NV first), also at $n = 400$. The reward of Eq. (1) is a learning signal only: one saved vehicle is worth $\lambda = 5$ points, i.e. as much as a 5% reduction of the initial distance. Unlike $f$, the reward therefore does not enforce the vehicle-first order: it is a shaping choice, and Table S15 varies $\lambda$.

TABLE S1
SOLOMON BENCHMARK ($n = 100$, $T = 5{,}000$, 5 SEEDS)

| Class | BKS | | ALNS-Uni | | ALNS-RW | | SG-ALNS | | Tuned static | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CNV | CTD | CNV | CTD | CNV | CTD | CNV | CTD | CNV | CTD |
| C1 | 90 | 7,455 | **90.0** | **7,455** | **90.0** | **7,455** | **90.0** | **7,455** | **90.0** | **7,455** |
| C2 | 24 | 4,719 | **24.0** | 4,771 | **24.0** | 4,769 | **24.0** | **4,719** | **24.0** | 4,769 |
| R1 | 143 | 14,524 | 149.8 | 14,496 | 149.6 | 14,416 | 148.2 | 14,530 | **148.0** | **14,549** |
| R2 | 30 | 10,461 | 33.2 | 10,469 | 33.4 | 10,396 | **33.0** | **10,439** | 33.4 | 10,406 |
| RC1 | 92 | 11,073 | 97.0 | 11,051 | 97.4 | 10,978 | **95.8** | **11,005** | 96.2 | 10,970 |
| RC2 | 26 | 8,954 | **27.0** | 9,110 | **27.0** | 9,100 | **27.0** | **9,040** | **27.0** | 9,041 |
| Total | 405 | 57,187 | 421.0 | 57,352 | 421.4 | 57,115 | **418.0** | **57,188** | 418.6 | 57,190 |
| Gap TD (%)[a] | – | | 2.09 | | 1.86 | | **1.48** | | 1.69 | |
| Primal integral (%) | – | | 5.44 | | 5.58 | | **5.05** | | 5.07 | |
| Time (s) | – | | 1.10 | | 1.09 | | 2.15 | | 0.97 | |

[a] mean over runs whose NV equals the reference NV.

CNV/CTD: vehicles and distance summed over the instances of a class (mean over seeds). Bold: best among the compared methods (lexicographically: CNV, then CTD).

TABLE S2
GREEDY ACTIONS OF THE TRAINED POLICIES ON SOLOMON ($T = 5{,}000$

| Model | $a^1$ (operators) | $a^2$ ($q$) | $a^3$ ($m$) | Share | # |
|---|---|---|---|---|---|
| SG-ALNS | RW weights | 30% | 3 | 1.00 | 1 |
| w/o expert | Worst/Regret-3 | 30% | 3 | 1.00 | 1 |
| w/o $a^1$ | RW weights | 20% | 1 | 0.73 | 2 |
| w/o $a^2$ | Worst/Regret-3 | $\mathcal{U}[5, 30]\%$ | 3 | 1.00 | 1 |
| w/o $a^3$ | Worst/Regret-3 | 30% | 1 | 1.00 | 1 |
| w/o dyn. TW | Worst/Regret-3 | 30% | 0.3 | 1.00 | 1 |
| MLP encoder | Worst/Regret-3 | 30% | 0.3 | 1.00 | 1 |
| SG-ALNS, seed 1 | Worst/Regret-3 | 30% | 0.3 | 1.00 | 1 |
| SG-ALNS, seed 2 | Wait/Regret-3 | 30% | 1 | 1.00 | 1 |
| SG-ALNS-C | Worst/Regret-3 | $\mathcal{U}[5, 30]\%$ | 0.3 | 1.00 | 1 |
| Iteration-level GNN | Worst/Regret-3 | $\mathcal{U}[5, 30]\%$ | 1 | 1.00 | 1 |
| Iteration-level MLP | Worst/Regret-3 | $\mathcal{U}[5, 30]\%$ | 1 | 1.00 | 1 |

All decisions of all runs). *Share*: fraction of decisions equal to the most frequent action; #: number of distinct actions observed.

TABLE S3
TRAINING SEEDS OF SG-ALNS (SOLOMON, $T = 5{,}000$)

| Seed | CNV | Gap % | PI % | vs. ALNS-RW | Most frequent action | Share |
|---|---|---|---|---|---|---|
| 0 | 418.0 | 1.48 | 5.05 | $-3.4$ (0.012) | RW weights, $q$=30%, $m$=3 | 1.00 |
| 1 | 420.2 | 2.05 | 5.24 | $-1.2$ (0.136) | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 |
| 2 | 418.8 | 2.11 | 4.94 | $-2.6$ (0.041) | Wait/Regret-3, $q$=30%, $m$=1 | 1.00 |
| 3 | 419.2 | 1.57 | 5.15 | $-2.2$ (0.058) | Worst/Regret-3, $q$=30%, $m$=1 | 0.60 |
| 4 | 420.2 | 1.64 | 5.20 | $-1.2$ (0.206) | Worst/Regret-3, $q$=30%, $m$=1 | 1.00 |

5 evaluation seeds. vs. ALNS-RW: $\Delta$CNV and two-sided Wilcoxon $p$ on per-instance vehicles; last columns: greedy action chosen most often and its share.

## S0 TABULAR RESULTS

*Statistics.* The paper bases every conclusion on hierarchical, two-sided tests (fleet first, then distance on equal-fleet instances) with TOST equivalence (Table IV of the paper). Tables S7, S9 and S10 additionally list W/T/L on $f$ with one-sided Wilcoxon $p$-values; these are descriptive and are not used for any conclusion.

Tables S1, S2 and S3 give the class-wise Solomon results, the greedy actions of all policies and the training-seed study that are shown graphically in Fig. 3 of the paper and Fig. S3.

## S1 HEADROOM AND STATIC CONFIGURATIONS

Table S4 compares static choices with ALNS-RW on generated instances before training: no static choice dominates ALNS-RW, which motivates the expert-inclusive action space. Table S5 lists the static-configuration grid used to select the tuned static baseline, and Table S6 the dynamic-headroom study (two-phase schedules).

TABLE S4
HEADROOM ANALYSIS

| Configuration | $\Delta$NV | $\Delta$TD (%) |
|---|---|---|
| ALNS-RW (reference) | +0 | +0.00 |
| Lead pair String/Regret-3 | -6 | +0.09 |
| RW operators, fixed $q = 30\%$ | -3 | +0.12 |
| RW operators, fixed $q = 20\%$ | -3 | +0.13 |
| Lead pair Shaw/Regret-3 | -2 | +0.23 |
| Lead pair Worst/Regret-3 | -2 | +0.34 |
| RW operators, fixed $q = 10\%$ | +4 | +1.41 |
| RW operators, fixed $q = 5\%$ | +11 | +2.17 |
| Lead pair Route/Noise | +6 | +2.28 |
| Lead pair Route/Greedy | +0 | +3.01 |

Static configurations vs. ALNS-RW on 24 generated instances ($n = 100$, $T = 2{,}500$, 3 seeds). $\Delta$NV: vehicles summed over 72 runs; $\Delta$TD: mean relative distance difference (positive = worse than RW).

TABLE S5
STATIC-CONFIGURATION GRID (30 CONFIGURATIONS) ON 24 GENERATED TUNING INSTANCES ($n = 100$, $T = 5{,}000$, 3 SEEDS)

| Static configuration | Score | $\Delta$NV |
|---|---|---|
| String/Regret-3, $q$=30%, $m$=0.3$^\dagger$ | +0.70 | -8 |
| String/Regret-3, $q$=30%, $m$=1 | +0.63 | -5 |
| String/Regret-3, $q$=20%, $m$=1 | +0.55 | -4 |
| String/Regret-3, $q$=$\mathcal{U}[5, 30]$%, $m$=1 | +0.49 | -8 |
| RW weights, $q$=30%, $m$=0.3 | +0.47 | -2 |
| ... | | |
| RW weights, $q$=30%, $m$=3$^\ddagger$ | +0.36 | -5 |
| ALNS-RW (reference) | +0.00 | +0 |
| RW weights, $q$=5%, $m$=0.3 | -2.88 | +10 |
| RW weights, $q$=5%, $m$=3 | -3.06 | +8 |

No benchmark instance is used. Score: $5\Delta\text{NV} + 100\Delta\text{TD}/\text{TD}$ w.r.t. ALNS-RW (higher is better), i.e. the reward weighting. †: selected as *Tuned static*; ‡: the constant action SG-ALNS converged to.

TABLE S6
DYNAMIC HEADROOM

| Schedule | Score |
|---|---|
| String/Regret-3, 20%, $m$=0.3 $\rightarrow$ String/Regret-3, 30%, $m$=3 | +0.81 |
| String/Regret-3, 30%, $m$=0.3 $\rightarrow$ RW weights, 30%, $m$=1 | +0.81 |
| String/Regret-3, 20%, $m$=0.3 $\rightarrow$ RW weights, 30%, $m$=3 | +0.79 |
| String/Regret-3, 20%, $m$=0.3 $\rightarrow$ String/Regret-3, 30%, $m$=1 | +0.79 |
| String/Regret-3, 20%, $m$=1 $\rightarrow$ RW weights, 30%, $m$=1 | +0.78 |
| String/Regret-3, 30%, $m$=0.3 (constant) | +0.70 |
| String/Regret-3, 30%, $m$=1 (constant) | +0.63 |
| String/Regret-3, 20%, $m$=1 (constant) | +0.55 |

Best two-phase schedules (first half $\rightarrow$ second half of the main phase) and best constants among the 10 best static actions, on the 24 generated tuning instances (3 seeds, $T = 5{,}000$). Score as in Table S5.

## S2 ABLATIONS

Each ablation is a separately trained policy (seed 0). Because every greedy policy is (almost) state-independent (Fig. 3a of the paper), the differences reflect the constant configuration each training run converged to, not the value of the removed component.

## S3 RUN-TIME BREAKDOWN AND CONVERGENCE

Table S8 splits the run time into the vehicle-reduction phase, the ALNS main phase and the policy overhead (features and inference). The policy accounts for 3–4% of the run time at $n = 400$; the rest is ALNS time, which depends on the operators and temperature selected. Figure S1 shows convergence on Solomon, and Fig. S2 the effect of the budget.

TABLE S7
ABLATIONS ON SOLOMON ($T = 5{,}000$, 5 SEEDS), EACH A SEPARATELY TRAINED POLICY (SAME BUDGET AND SEED)

| Method | CNV | CTD | Gap (%) | PI (%) | Time (s) | W/T/L | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|---|
| SG-ALNS | 418.0 | 57,188 | 1.48 | 5.05 | 2.15 | – | – |
| w/o expert actions & prior | 419.0 | 57,017 | 1.40 | 5.20 | 2.47 | 18/16/22 | 0.657 |
| w/o $a^1$ (RW operators) | 421.6 | 57,142 | 1.85 | 5.65 | 1.37 | 27/16/13 | 0.002 |
| w/o $a^2$ (default $q$ range) | 420.6 | 57,269 | 1.95 | 5.32 | 1.54 | 26/16/14 | 0.002 |
| w/o $a^3$ (scheduled $\tau$) | 420.2 | 57,077 | 1.64 | 5.20 | 2.46 | 23/16/17 | 0.062 |
| w/o dynamic TW features | 420.2 | 57,281 | 2.05 | 5.24 | 2.46 | 28/16/12 | 0.002 |
| MLP encoder (no message passing) | 420.2 | 57,281 | 2.05 | 5.24 | 2.35 | 28/16/12 | 0.002 |

Columns as in Table III. Because every greedy policy is state-independent (Table S2), these differences reflect the constant configuration each training run converged to.

TABLE S8
WALL-CLOCK BREAKDOWN PER RUN (SECONDS)

| Set | Method | $T$ | Reduction | ALNS | Policy | Total |
|---|---|---|---|---|---|---|
| Solomon | ALNS-RW | 10,000 | 0.13 | 1.91 | 0.00 | 2.04 |
| Solomon | Tuned ALNS-RW | 10,000 | 0.12 | 2.59 | 0.00 | 2.71 |
| Solomon | Tuned static | 10,000 | 0.12 | 2.40 | 0.00 | 2.57 |
| HG-400 | ALNS-RW | 5,000 | 4.86 | 12.69 | 0.00 | 17.55 |
| HG-400 | Tuned ALNS-RW | 5,000 | 4.77 | 17.66 | 0.00 | 22.42 |
| HG-400 | Tuned static | 5,000 | 4.80 | 16.09 | 0.00 | 21.34 |
| HG-400 | SG-ALNS | 5,000 | 4.68 | 23.26 | 0.96 | 29.30 |
| HG-400 | SG-ALNS-C | 5,000 | 4.69 | 17.77 | 0.91 | 23.76 |

Vehicle-reduction phase, ALNS main phase, and policy overhead (feature construction and network inference).

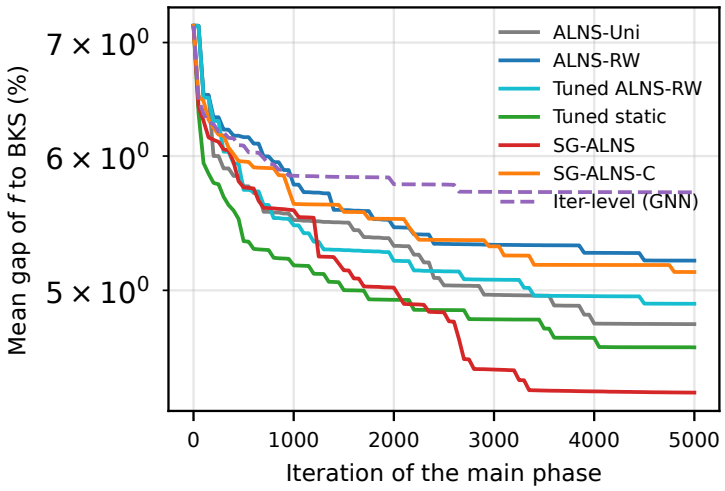



Fig. S1. Mean gap of the best-so-far objective $f$ to the BKS along the main phase (Solomon, 3 seeds, log scale).

## S4 TIME-MATCHED COMPARISON AND BUDGETS

## S5 ACTIONS OF SG-ALNS-C

## S6 INSTANCE SIZE

## S7 LEARNED ACTIONS

Table S13 lists the greedy action of every trained policy; the six controls are summarised in Table I of the paper.

## S8 BKS CROSS-CHECK AND REPRODUCIBILITY

The Solomon BKS table was cross-checked against the 56 CVRPLIB solutions (DIMACS convention: distance-only objective with distances truncated to one decimal). Re-evaluated with unrounded distances, 48 of them are feasible; on the 19 instances where they use the BKS fleet, they reproduce the BKS distance exactly. Over 8,232 Solomon runs, no solution dominates a

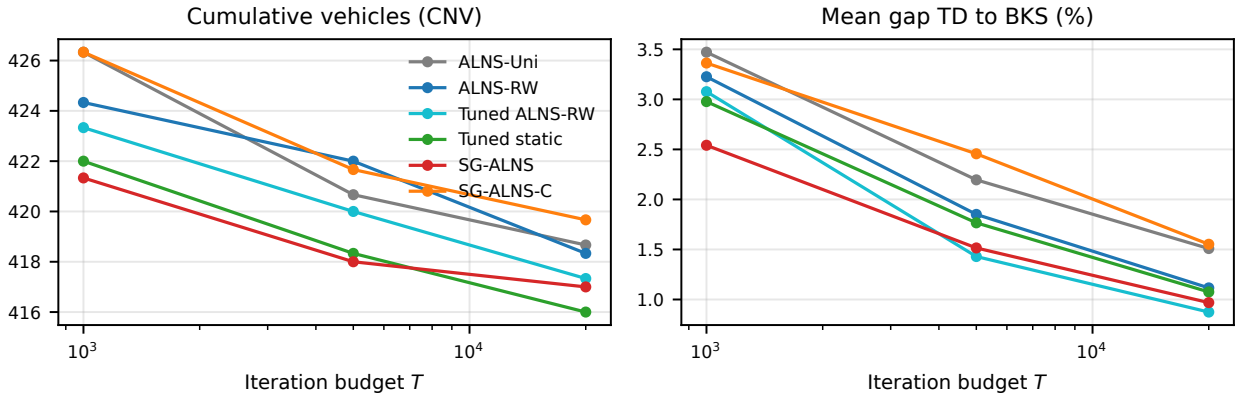



Fig. S2. Cumulative vehicles and TD gap on Solomon as a function of the budget $T$ (3 seeds).

TABLE S9
TIME-MATCHED COMPARISON ON SOLOMON (5 SEEDS)

| Method | $T$ | Time (s) | CNV | CTD | Gap (%) |
|---|---|---|---|---|---|
| SG-ALNS | 5,000 | 2.15 | 418.0 | 57,188 | 1.48 |
| SG-ALNS-C | 5,000 | 1.71 | 422.0 | 57,512 | 2.44 |
| ALNS-RW | 10,000 | 2.04 | 419.8 | 56,987 | 1.40 |
| Tuned ALNS-RW | 10,000 | 2.71 | 418.8 | 56,931 | 1.25 |
| Tuned static | 10,000 | 2.57 | 417.8 | 56,990 | 1.28 |

The baselines receive twice the iterations, which roughly equalises wall-clock time with SG-ALNS at $T = 5{,}000$.

TABLE S10
ITERATION BUDGET ON SOLOMON (3 SEEDS)

| | CNV | | | | | Gap TD (%) | | | | | SG-ALNS vs. | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | RW | RWt | St | SG | SG-C | RW | RWt | St | SG | SG-C | RWt | St |
| 1,000 | 424.3 | 423.3 | 422.0 | 421.3 | 426.3 | 3.23 | 3.08 | 2.98 | 2.54 | 3.36 | 25/14/17 (0.002) | 27/14/15 (0.028) |
| 5,000 | 422.0 | 420.0 | 418.3 | 418.0 | 421.7 | 1.85 | 1.43 | 1.77 | 1.52 | 2.46 | 24/17/15 (0.064) | 21/16/19 (0.150) |
| 20,000 | 418.3 | 417.3 | 416.0 | 417.0 | 419.7 | 1.12 | 0.87 | 1.08 | 0.97 | 1.55 | 21/17/18 (0.506) | 16/16/24 (0.753) |

CNV and gap for ALNS-RW, tuned ALNS-RW (RWt), tuned static (St), SG-ALNS (SG) and SG-ALNS-C (SG-C); last columns: SG-ALNS vs. tuned ALNS-RW and vs. tuned static (W/T/L, one-sided Wilcoxon $p$). Tuned baselines were tuned for $T = 5{,}000$; SG-ALNS was trained with $T = 2{,}500$ and SG-ALNS-C with $T \in \{1{,}000, 2{,}500, 5{,}000\}$.

BKS, and 25 BKS are reproduced exactly. Gehring–Homberger instances with $n = 400$ were converted from the VRPLIB files of the PyVRP instance repository; the same converter reproduces our $n = 200$ files exactly. The script `run_all.sh` regenerates every result, and `make_paper.sh` every number, table and figure of the paper.

## S9 ADDITIONAL ANALYSES

The state-sensitivity probe of the trained policies is Table V of the paper. *How much tuning is needed?* Table S14 selects constant actions on half of the generated tuning instances and scores them on the other half. *Reward.* Table S15 retrains seed 0 with other vehicle weights $\lambda$. *Iteration-level agents near their horizon.* Table S16 evaluates them at $T = 200$. *Instance classes at* $n = 400$. Table S17 splits the $n = 400$ results by class.

## S10 ADDITIONAL FIGURES

TABLE S11
GREEDY ACTIONS OF SG-ALNS-C BY INSTANCE SIZE $n$ AND BUDGET $T$

| $n$ | $T$ | Mode, first half | Mode, second half | Share | # | $\bar{q}$ (%) |
|---|---|---|---|---|---|---|
| 100 | 1,000 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | 1.00 | 1 | 17.5 |
| 100 | 5,000 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | 1.00 | 1 | 17.5 |
| 100 | 20,000 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | 1.00 | 1 | 17.5 |
| 200 | 5,000 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | 1.00 | 1 | 17.5 |
| 400 | 5,000 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | Worst/Regret-3, $\mathcal{U}[5,30]\%$, 0.3 | 1.00 | 1 | 17.5 |

Most frequent action in the first and second half of the run, share of the most frequent action over the run, number of distinct actions, and mean destroy fraction ($\mathcal{U}[5,30]\%$ counted as 17.5%).

TABLE S12
GEHRING–HOMBERGER INSTANCES WITH $n = 200$ AND $n = 400$ ($T = 5{,}000$, 3 SEEDS, 60 INSTANCES EACH)

| $n$ | Method | CNV | CTD | Gap (%) | PI (%) | Time (s) | vs. ALNS-RW |
|---|---|---|---|---|---|---|---|
| 200 | ALNS-RW | 697.7 | 173,654 | 1.93 | 0.64 | 2.92 | – |
| | Tuned ALNS-RW | 697.7 | 173,225 | 1.66 | 0.62 | 6.32 | 34/6/20 (0.012) |
| | Tuned static | 697.7 | 173,741 | 1.93 | 0.53 | 3.43 | 32/4/24 (0.120) |
| | SG-ALNS | 697.3 | 174,722 | 2.41 | 0.65 | 5.28 | 18/6/36 (0.979) |
| | SG-ALNS-C | 698.0 | 172,937 | 1.66 | 0.56 | 6.97 | 34/6/20 (0.016) |
| 400 | ALNS-RW | 1397.3 | 411,122 | 2.79 | 0.66 | 17.55 | – |
| | Tuned ALNS-RW | 1397.0 | 411,267 | 2.75 | 0.63 | 22.42 | 35/0/25 (0.327) |
| | Tuned static | 1395.7 | 417,026 | 3.79 | 0.47 | 21.34 | 27/1/32 (0.844) |
| | SG-ALNS | 1396.0 | 422,567 | 5.01 | 0.61 | 29.30 | 17/1/42 (1.000) |
| | SG-ALNS-C | 1396.7 | 405,283 | 1.53 | 0.51 | 23.76 | 43/0/17 (<0.001) |

All policies and configurations were obtained with $n \leq 200$ (SG-ALNS-C) or $n \leq 100$ (all others). Gap and PI relative to the best solution found in the experiment; last column: W/T/L vs. ALNS-RW (one-sided Wilcoxon).

TABLE S13
GREEDY ACTIONS OF THE TRAINED POLICIES ON SOLOMON

| Policy | Most frequent action | Share | # | Switch./run | CNV |
|---|---|---|---|---|---|
| PPO, seed 0 | RW weights, $q$=30%, $m$=3 | 1.00 | 1 | 0.0 | 418.0 |
| PPO, seed 1 | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, seed 2 | Wait/Regret-3, $q$=30%, $m$=1 | 1.00 | 1 | 0.0 | 418.8 |
| PPO, seed 3 | Worst/Regret-3, $q$=30%, $m$=1 | 0.60 | 2 | 6.6 | 419.2 |
| PPO, seed 4 | Worst/Regret-3, $q$=30%, $m$=1 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, SG-ALNS-C | Worst/Regret-3, $q$=$\mathcal{U}$, $m$=0.3 | 1.00 | 1 | 0.0 | 422.0 |
| PPO, neutral prior, seed 0 | Worst/Regret-3, $q$=$\mathcal{U}$, $m$=0.3 | 0.78 | 2 | 1.7 | 422.6 |
| PPO, w/o expert | Worst/Regret-3, $q$=30%, $m$=3 | 1.00 | 1 | 0.0 | 419.0 |
| PPO, w/o $a^1$ | RW weights, $q$=20%, $m$=1 | 0.73 | 2 | 8.7 | 421.6 |
| PPO, w/o $a^2$ | Worst/Regret-3, $q$=$\mathcal{U}$, $m$=3 | 1.00 | 1 | 0.0 | 420.6 |
| PPO, w/o $a^3$ | Worst/Regret-3, $q$=30%, $m$=1 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, w/o dyn. TW | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, MLP encoder | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, $\lambda = 1$ | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, $\lambda = 25$ | Worst/Regret-3, $q$=30%, $m$=1 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, neutral prior, seed 1 | Random/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.6 |
| PPO, neutral prior, seed 2 | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 420.2 |
| PPO, seed 0, 5× training | Shaw/Regret-3, $q$=30%, $m$=0.3 | 1.00 | 1 | 0.0 | 419.6 |
| Iteration-level GNN* | Worst/Regret-3, $q$=$\mathcal{U}$, $m$=1 | 1.00 | 1 | – | – |
| Iteration-level MLP* | Worst/Regret-3, $q$=$\mathcal{U}$, $m$=1 | 1.00 | 1 | – | – |

$T = 5{,}000$, all decisions of all runs: most frequent action, its share, number of distinct actions, mean number of action switches per run, and CNV. *One decision per iteration, measured on 14 Solomon instances (`iter_actions.csv`).

TABLE S14
TUNING BUDGET VS. HELD-OUT QUALITY

| $b$ | CPU-min | Held-out score | P10 | P90 |
|---|---|---|---|---|
| 1 | 1 | $-1.42$ | $-3.18$ | +0.34 |
| 3 | 2 | $-0.44$ | $-1.45$ | +0.68 |
| 10 | 5 | +0.15 | $-0.57$ | +0.78 |
| 30 | 16 | +0.47 | +0.06 | +0.83 |
| 100 | 53 | +0.47 | +0.18 | +0.83 |
| 285 | 150 | +0.18 | +0.18 | +0.18 |

285 constant actions, generated instances, $T = 5{,}000$, 3 seeds. For a budget of $b$ random configurations, the best one on 12 selection instances is scored on 12 held-out instances (score vs. ALNS-RW: $5\Delta$NV$+100\Delta$TD/TD; mean, 10th and 90th percentile over 500 draws). CPU-min: cost of the selection runs.

TABLE S15
REWARD SENSITIVITY

| $\lambda$ | CNV | Gap % | vs. ALNS-RW | Most frequent action | Share |
|---|---|---|---|---|---|
| 1 | 420.2 | 2.05 | $-1.2$ (0.136) | Worst/Regret-3, $q$=30%, $m$=0.3 | 1.00 |
| 5 (paper) | 418.0 | 1.48 | $-3.4$ (0.012) | RW weights, $q$=30%, $m$=3 | 1.00 |
| 25 | 420.2 | 1.64 | $-1.2$ (0.206) | Worst/Regret-3, $q$=30%, $m$=1 | 1.00 |

Seed 0, Solomon, $T = 5{,}000$, 5 seeds; $\lambda$: weight of a saved vehicle in Eq. (1). vs. ALNS-RW: $\Delta$CNV and two-sided Wilcoxon $p$; last columns: most frequent greedy action and its share.

TABLE S16
SHORT BUDGET $T = 200$

| Method | CNV | Gap % | Time s | vs. ALNS-RW |
|---|---|---|---|---|
| ALNS-RW | 428.3 | 5.86 | 0.14 | – |
| Tuned static | 425.3 | 4.59 | 0.17 | $-3.0$ (0.009) |
| SG-ALNS | 425.0 | 5.37 | 0.20 | $-3.3$ (0.026) |
| Iteration-level GNN | 428.3 | 4.83 | 0.58 | $-0.0$ (0.750) |
| Iteration-level MLP | 428.3 | 4.83 | 0.38 | $-0.0$ (0.750) |

Solomon, 3 seeds; close to the 100-iteration training episodes of the iteration-level agents. vs. ALNS-RW: $\Delta$CNV and two-sided Wilcoxon $p$.

TABLE S17
GEHRING–HOMBERGER $n = 400$ PER INSTANCE CLASS

| Method | C1 | C2 | R1 | R2 | RC1 | RC2 |
|---|---|---|---|---|---|---|
| ALNS-RW | 381.0 / 1.69 | 120.0 / 1.44 | 364.0 / 3.47 | 80.0 / 1.64 | 362.0 / 5.91 | 90.3 / 2.88 |
| Tuned ALNS-RW | 380.7 / 2.00 | 120.0 / 1.26 | 364.0 / 3.31 | 80.0 / 1.59 | 362.0 / 6.01 | 90.3 / 2.63 |
| Tuned static | 380.7 / 5.21 | 120.0 / 0.95 | 364.0 / 5.39 | 80.0 / 1.44 | 361.3 / 8.24 | 89.7 / 1.97 |
| SG-ALNS | 380.7 / 6.30 | 120.0 / 2.28 | 364.0 / 5.18 | 80.0 / 2.58 | 361.3 / 10.17 | 90.0 / 3.57 |
| SG-ALNS-C | 381.0 / 1.04 | 120.0 / 0.51 | 364.0 / 0.92 | 80.0 / 1.75 | 361.3 / 1.26 | 90.3 / 3.70 |

10 instances each, 3 seeds: cumulative vehicles / mean TD gap (%) to the best solution found, over runs with that fleet.

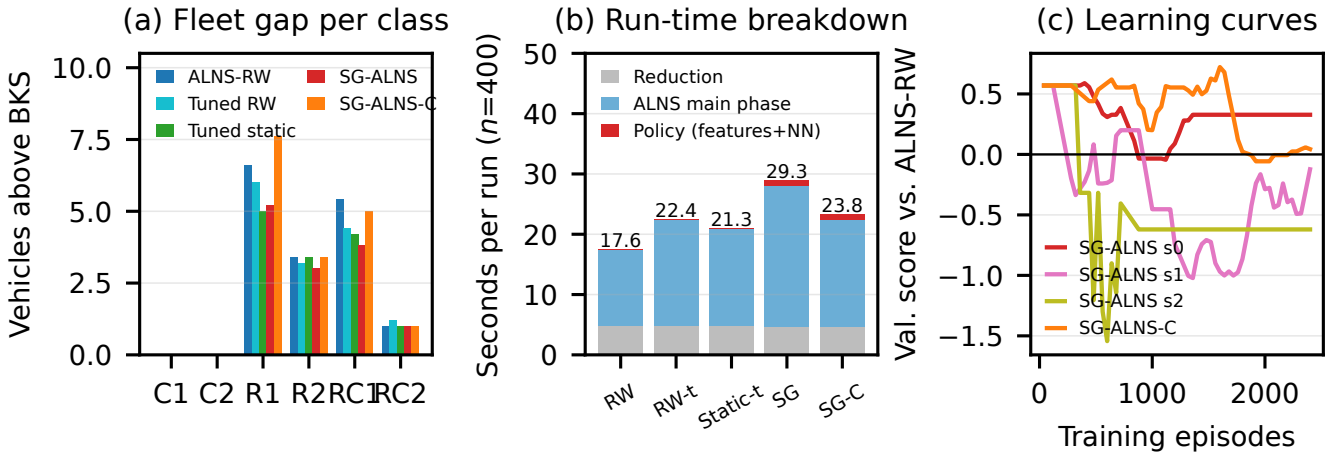



Fig. S3. (a) Vehicles above the BKS per Solomon class (sum over instances, mean over 5 seeds). (b) Mean run time per instance at $n = 400$, split into vehicle reduction, ALNS main phase and policy overhead (number: total). (c) Validation score against ALNS-RW on 24 generated instances during training (rolling mean of 4 evaluations).

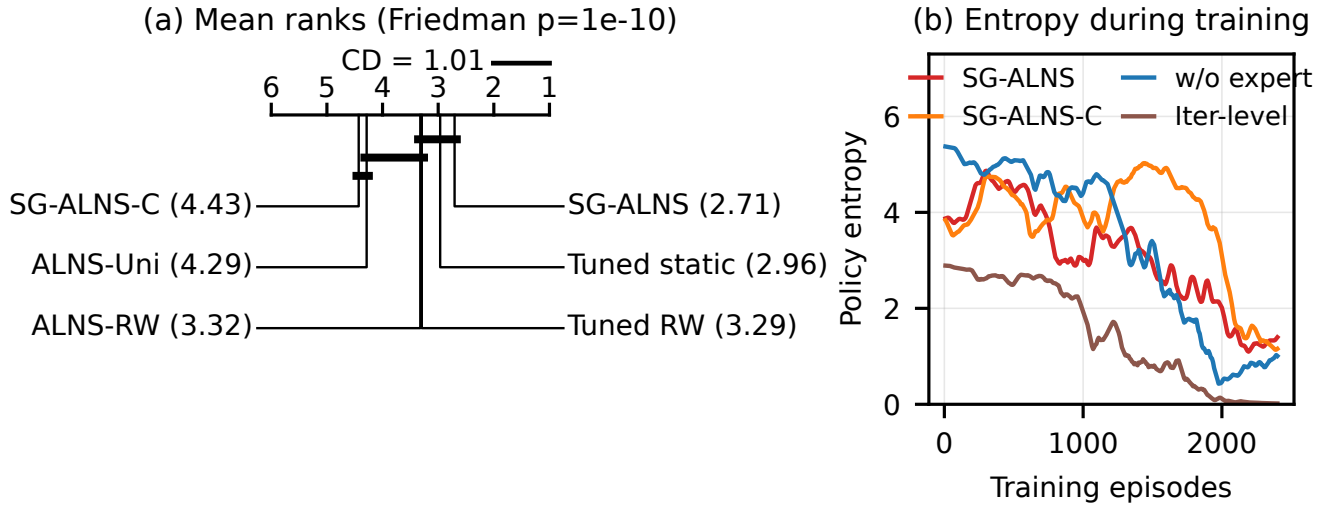



Fig. S4. (a) Critical-difference diagram (Friedman–Nemenyi, $\alpha = 0.05$) of the six methods run with 5 seeds on Solomon, $T = 5{,}000$; methods joined by a bar are not significantly different. (b) Policy entropy (sum over the three heads) during training.